

```
from google.colab import files
uploaded = files.upload()
```

Screenshot ... 142020.png

Screenshot 2026-07-13 142020.png(image/png) - 268920 bytes, last modified: 13/07/2026 - 100% done
Saving Screenshot 2026-07-13 142020.png to Screenshot 2026-07-13 142020 (1).png

```
import cv2
import matplotlib.pyplot as plt

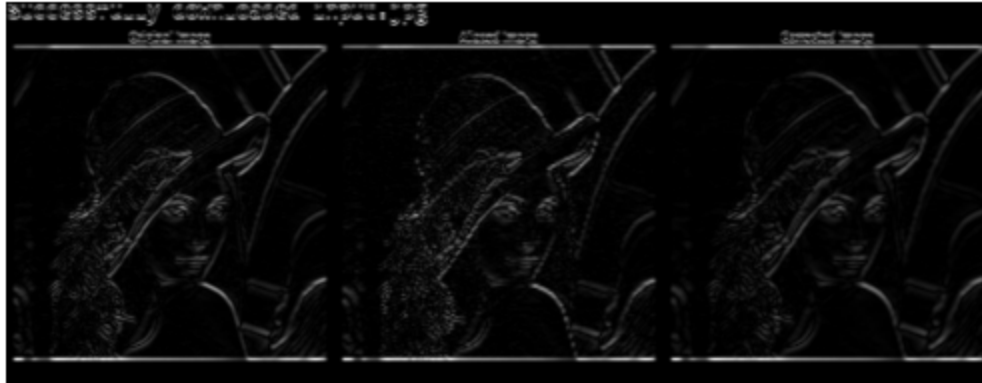
# Try loading the image without forcing grayscale first
img_color = cv2.imread('/content/Screenshot 2026-07-13 142020 (1).png')

if img_color is None:
    print("Error: Could not load image from the specified path.")
else:
    # Convert to grayscale if it's a color image
    if len(img_color.shape) == 3:
        img = cv2.cvtColor(img_color, cv2.COLOR_BGR2GRAY)
    else: # Already grayscale or 1-channel
        img = img_color

    sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

    plt.imshow(abs(sobel_y), cmap='gray')
    plt.title('Sobel Edge Detection - Y Axis')
    plt.axis('off')
    plt.show()
```

Sobel Edge Detection - Y Axis



Cause of Aliasing:

- Aliasing occurs when the image is sampled below the Nyquist rate.
- High-frequency details appear as jagged edges or moiré patterns.
- Quantization reduces pixel intensity resolution but is not the main cause of

Consecutive Approach:

- Apply an anti-aliasing (Gaussian) filter before downsampling.
- Use better interpolation methods such as INTER_AREA or INTER_LANCZOS4.
- Because the same line operation is possible